HAMID KHABIRI, PhD

ROBOTICS ENGINEER | MECHANICAL DESIGN · LOW-LEVEL CONTROL · SYSTEM INTEGRATION

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PROFESSIONAL SUMMARY

Robotics engineer with 8+ years of research and industry experience independently developing robotic systems from concept to functional hardware. Combines mechanical design and fabrication with real-time actuator control, model-based control and safety validation.

CORE TECHNICAL SKILLS

Mechanical design & manufacturing: SolidWorks; mechanism/transmission design; actuator selection; engineering calculations, tolerances, DFM, CNC drawings and 3D printing.

Robotics control & modeling: PID tuning; joint-/Cartesian-space control (impedance/admittance); robot kinematics/dynamics (FK/IK, Jacobians); trajectory planning and model-based feedforward.

Real-time control & software: TwinCAT 3 (Structured Text, PLC/NC, HMI); EtherCAT (PDO/SDO, timing); C++/MFC, and Unity.

Electromechanical integration & safety: BLDC motors and servo drives; encoder and F/T sensor calibration; schematics, wiring, hardware bring-up and fault diagnosis; risk assessment; safety functions (emergency stop, STO, virtual walls).

ENGINEERING EXPERIENCE

Postdoctoral Fellow - Robotics R&D | University of Alberta • Nov 2024–Present

- Independently designed and built a custom 3-DOF robotic system: mechanism architecture, SolidWorks CAD, component sizing, engineering calculations, tolerances, DFM, fabrication and assembly.
- Selected and integrated three Maxon BLDC motors with EPOS4 EtherCAT drives, transmissions, absolute encoders and force/torque sensing; designed and wired electrical hardware and calibrated sensor feedback.
- Developed real-time low-level motion control in TwinCAT 3/EtherCAT (PLC/NC, PDO/SDO and synchronization); configured position/velocity/torque operation, tuned servo controllers in EPOS Studio and debugged fieldbus timing/faults.
- Implemented FK/IK, Jacobian-based control and trajectory planning in TwinCAT 3 (Structured Text); integrated impedance/admittance control and model-based feedforward.
- Developed TwinCAT HMI and Unity rehabilitation games; implemented layered hardware/software safety and fault handling; conducted structured bench verification and subsystem testing.

Mechanical Designer | Noavaran Sanat Rabin (NSR) • Apr 2023–Sep 2024

- Designed, fabricated and assembled a cell-culture incubator with real-time image monitoring; designed mechanical components of a stereotactic head frame, from machining planning to assembly.

Mechanical Engineer | Sina Robotics and Medical Innovators • Jan 2019–Dec 2020

- Designed, fabricated and assembled mechanisms for the SinaSim laparoscopic surgery simulator; completed engineering calculations and manufacturing drawings; tested, troubleshot and improved the hardware.

SELECTED ROBOTICS PROJECTS

Wearable robotic hand orthosis (PhD): Independently designed, fabricated and controlled an underactuated wearable robot; implemented a C++/MFC interface and performed clinical evaluation.

MRI-compatible wrist robot: Designed, fabricated and assembled a 2-DOF pneumatic wrist robot for fMRI research.

EDUCATION

PhD, Medical Robotics | Tehran University of Medical Sciences, Tehran, Iran — 2022

MSc, Mechanical Engineering (Biomechanics) | Sahand University of Technology, Tabriz, Iran — 2014

BSc, Biomedical Engineering (Biomechanics) | Islamic Azad University, Science and Research Branch, Tehran, Iran — 2011

PATENTS & RESEARCH

Inventor on published U.S. patent application US20260109022A1 and three Iranian patents; co-author of eight peer-reviewed journal articles.